

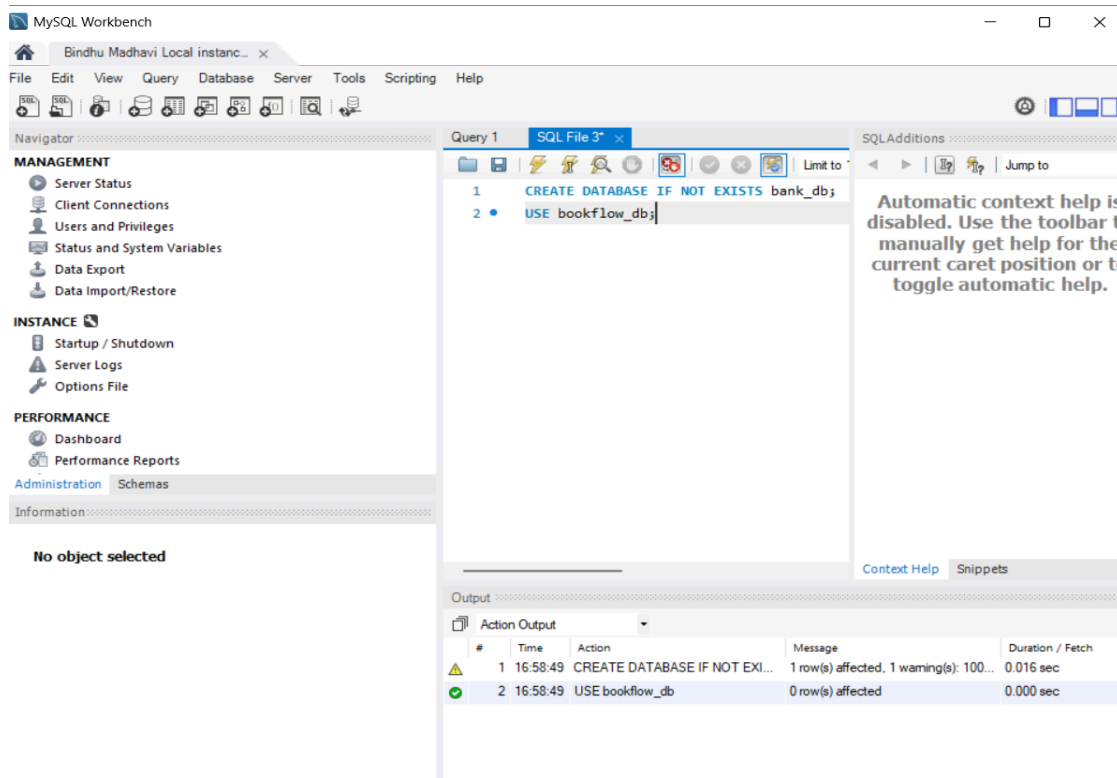
WEEK-2

Name: P. Bindhu Madhavi

Roll Number: 2520030077

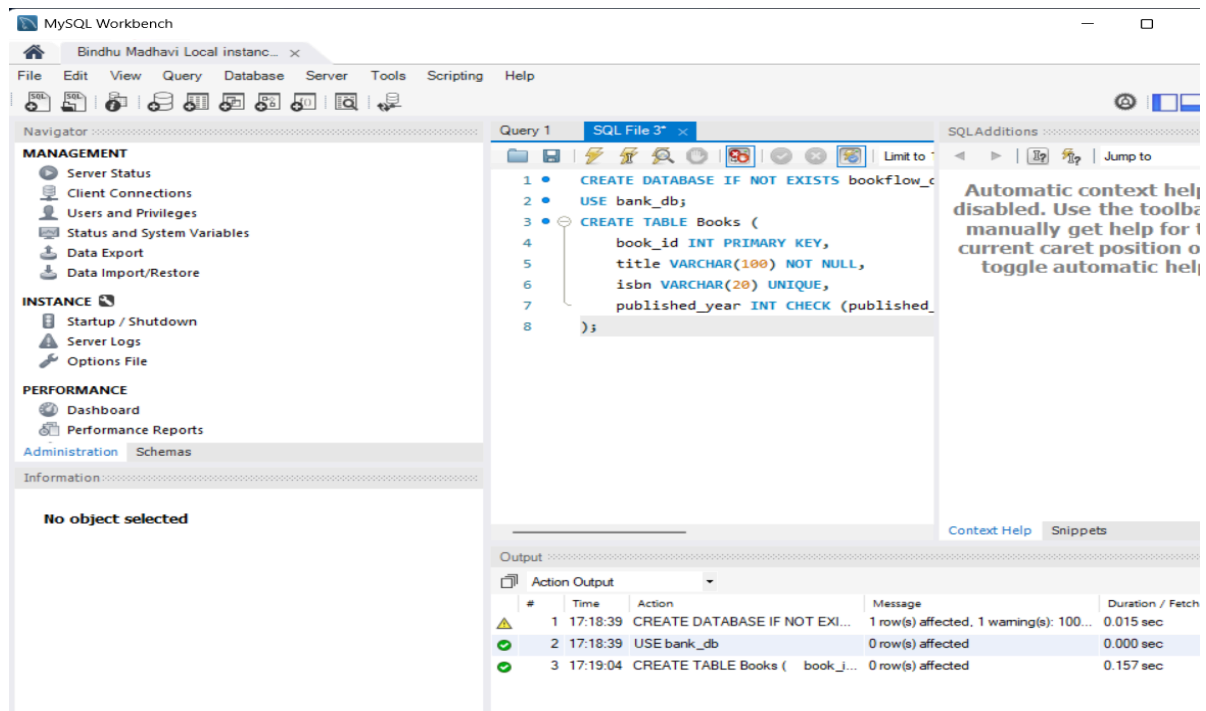
Step 1 – Create Database

Creates a new database to hold all the tables required for the Library Management System.



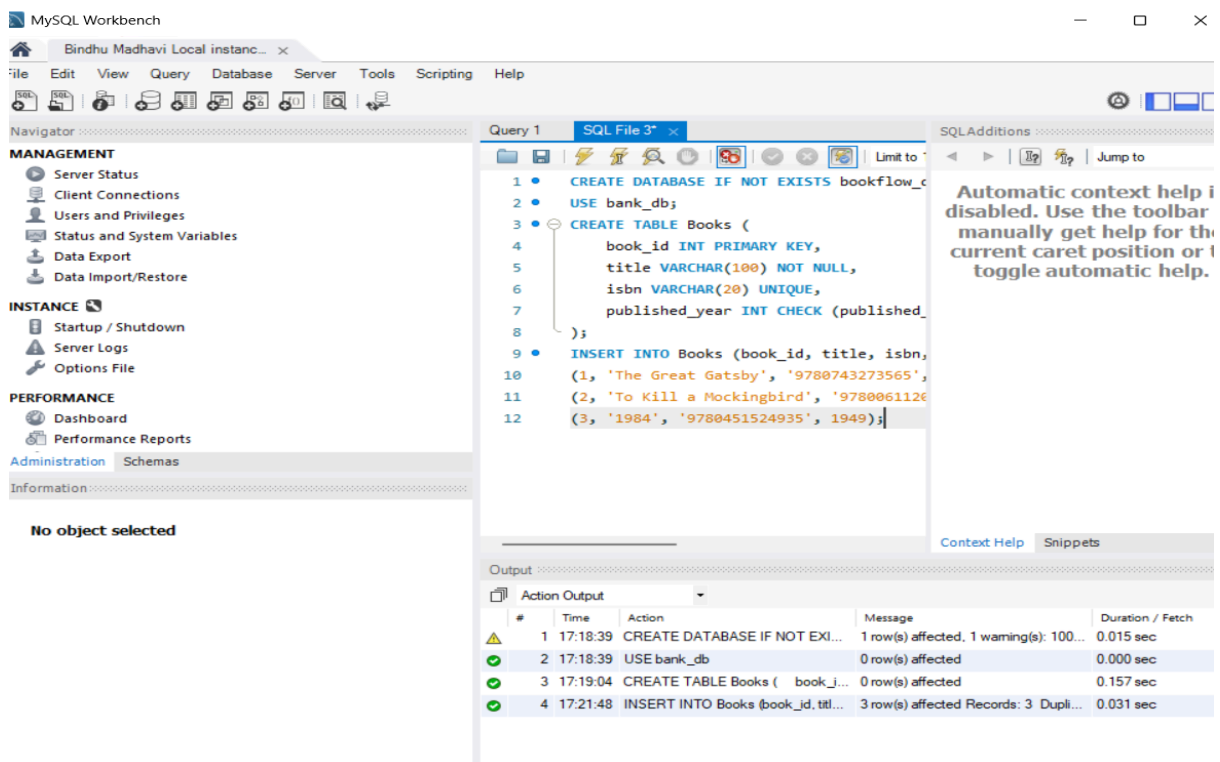
Step2-Create Books Table (DDL + Constraints)

Defines the Books table with a primary key, NOT NULL, and UNIQUE constraints to maintain data integrity.



Step 3 - Insert Books

Inserts sample book records into the Books table using INSERT statements.



Step 4 - Display Books

Retrieves and displays all records currently stored in the Books table.

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
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- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 SQL File 3*

```

7 published_year INT CHECK (published_
8 );
9 • INSERT INTO Books (book_id, title, isbn,
10 (1, 'The Great Gatsby', '9780743273565',
11 (2, 'To Kill a Mockingbird', '9780061120
12 (3, '1984', '9780451524935', 1949);
13 • SELECT * FROM Books;

```

Result Grid

	book_id	title	isbn
1		The Great Gatsby	9780743273565
2		To Kill a Mockingbird	9780061120084
3		1984	9780451524935
*	NULL	NULL	NULL

Books 1 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:18:39	CREATE DATABASE IF NOT EXI...	1 row(s) affected, 1 warning(s): 10...	0.015 sec
2	17:18:39	USE bank_db	0 row(s) affected	0.000 sec
3	17:19:04	CREATE TABLE Books (book...	0 row(s) affected	0.157 sec
4	17:21:48	INSERT INTO Books (book_id, tit...	3 row(s) affected Records: 3 Dupl...	0.031 sec
5	17:24:21	SELECT * FROM Books LIMIT 0, ...	3 row(s) returned	0.000 sec / 0.000

Automatic context help disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.

Step 5 - Create Members Table

Creates the Members table to store the personal and contact details of library members.

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
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- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 SQL File 3*

```

3 • CREATE TABLE Books (
4   book_id INT PRIMARY KEY,
5   title VARCHAR(100) NOT NULL,
6   isbn VARCHAR(20) UNIQUE,
7   published_year INT CHECK (published_
8 );
9 • INSERT INTO Books (book_id, title, isbn,
10 (1, 'The Great Gatsby', '9780743273565',
11 (2, 'To Kill a Mockingbird', '9780061120
12 (3, '1984', '9780451524935', 1949);
13 • SELECT * FROM Books;
14 • CREATE TABLE Members (
15   member_id INT PRIMARY KEY,
16   full_name VARCHAR(100),
17   email VARCHAR(100) UNIQUE
18 );

```

Books 1 x Apply Revert Context Help Snippets

Output

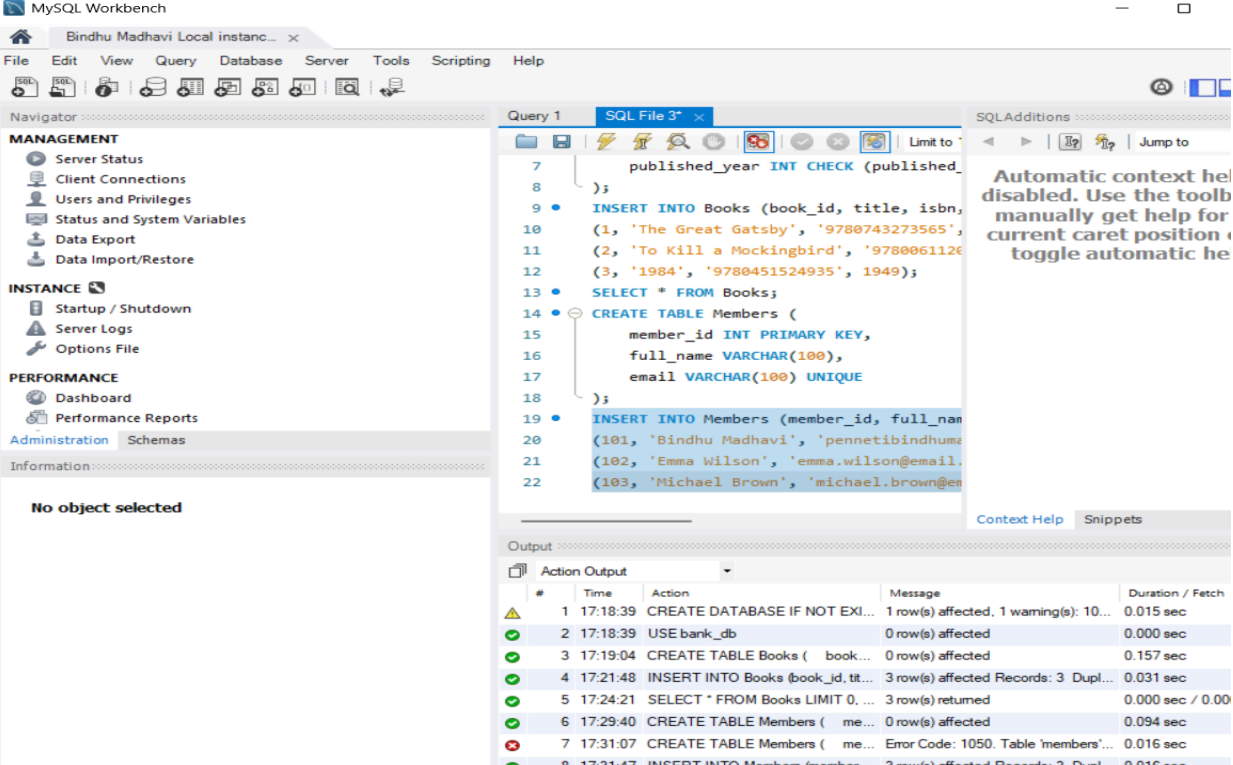
Action Output

#	Time	Action	Message	Duration / Fetch
1	17:18:39	CREATE DATABASE IF NOT EXI...	1 row(s) affected, 1 warning(s): 10...	0.015 sec
2	17:18:39	USE bank_db	0 row(s) affected	0.000 sec
3	17:19:04	CREATE TABLE Books (book...	0 row(s) affected	0.157 sec
4	17:21:48	INSERT INTO Books (book_id, tit...	3 row(s) affected Records: 3 Dupl...	0.031 sec
5	17:24:21	SELECT * FROM Books LIMIT 0, ...	3 row(s) returned	0.000 sec / 0.000
6	17:29:40	CREATE TABLE Members (me...	0 row(s) affected	0.094 sec

Automatic context help disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.

Step 6 - Insert Members

Adds sample member records into the Members table.



The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'Navigator' pane with sections for MANAGEMENT, INSTANCE, and PERFORMANCE. The main editor displays a SQL script with the following queries:

```
7      published_year INT CHECK (published_
8    );
9    • INSERT INTO Books (book_id, title, isbn,
10   (1, 'The Great Gatsby', '9780743273565',
11   (2, 'To Kill a Mockingbird', '9780061126
12   (3, '1984', '9780451524935', 1949));
13   • SELECT * FROM Books;
14   • CREATE TABLE Members (
15     member_id INT PRIMARY KEY,
16     full_name VARCHAR(100),
17     email VARCHAR(100) UNIQUE
18   );
19   • INSERT INTO Members (member_id, full_nam
20   (101, 'Bindhu Madhavi', 'pennetibindhume
21   (102, 'Emma Wilson', 'emma.wilson@email.
22   (103, 'Michael Brown', 'michael.brown@e
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	17:18:39	CREATE DATABASE IF NOT EXI...	1 row(s) affected, 1 warning(s): 10...	0.015 sec
2	17:18:39	USE bank_db	0 row(s) affected	0.000 sec
3	17:19:04	CREATE TABLE Books (book...	0 row(s) affected	0.157 sec
4	17:21:48	INSERT INTO Books (book_id, tit...	3 row(s) affected Records: 3 Dupl...	0.031 sec
5	17:24:21	SELECT * FROM Books LIMIT 0. ...	3 row(s) returned	0.000 sec / 0.00
6	17:29:40	CREATE TABLE Members (me...	0 row(s) affected	0.094 sec
7	17:31:07	CREATE TABLE Members (me...	Error Code: 1050. Table 'members'...	0.016 sec
8	17:31:47	INSERT INTO Members (member...	3 row(s) affected Records: 3 Dupl...	0.016 sec

Step 7 - Display Members

Displays all member records to verify that the data was inserted correctly.

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information: No object selected

Query 1 SQL File 3*

```

17 email VARCHAR(100) UNIQUE
18 );
19 INSERT INTO Members (member_id, full_name, email)
20 VALUES (101, 'Bindhu Madhavi', 'pennetibindhumadha@gmail.com'),
21 (102, 'Emma Wilson', 'emma.wilson@email.com'),
22 (103, 'Michael Brown', 'michael.brown@email.com');
23 SELECT * FROM Members;

```

Result Grid

member_id	full_name	email
101	Bindhu Madhavi	pennetibindhumadha@gmail.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

Members 2 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:18:39	CREATE DATABASE IF NOT EXISTS bank_db;	1 row(s) affected, 1 warning(s): 10056 (Warning: 1286: Cannot open file: 'data_dir' for table 'bank_db')	0.015 sec
2	17:18:39	USE bank_db	0 row(s) affected	0.000 sec
3	17:19:04	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100), author VARCHAR(100), year INT, price DECIMAL(10,2));	0 row(s) affected	0.157 sec
4	17:21:48	INSERT INTO Books (book_id, title, author, year, price) VALUES (1, 'The Great Gatsby', 'F. Scott Fitzgerald', 1925, 10.99), (2, '1984', 'George Orwell', 1949, 12.99), (3, 'The Catcher in the Rye', 'J.D. Salinger', 1951, 10.99);	3 row(s) affected Records: 3 Duplicates: 0 / 0.031 sec	
5	17:24:21	SELECT * FROM Books LIMIT 0, 3;	3 row(s) returned	0.000 sec / 0.000 sec
6	17:29:40	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	0 row(s) affected	0.094 sec
7	17:31:07	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	Error Code: 1050. Table 'members' already exists	0.016 sec
8	17:31:47	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Bindhu Madhavi', 'pennetibindhumadha@gmail.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'Michael Brown', 'michael.brown@email.com');	3 row(s) affected Records: 3 Duplicates: 0 / 0.016 sec	
9	17:33:11	SELECT * FROM Members LIMIT 0, 3;	3 row(s) returned	0.000 sec / 0.000 sec

Automatic context help disabled. Use the toolbar manually get help for current caret position or toggle automatic help.

Step 8 - Create Loans Table

Creates the Loans table with foreign keys referencing the Books and Members tables.

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information: No object selected

Query 1 SQL File 3*

```

16 full_name VARCHAR(100),
17 email VARCHAR(100) UNIQUE
18 );
19 INSERT INTO Members (member_id, full_name, email)
20 VALUES (101, 'Bindhu Madhavi', 'pennetibindhumadha@gmail.com'),
21 (102, 'Emma Wilson', 'emma.wilson@email.com'),
22 (103, 'Michael Brown', 'michael.brown@email.com');
23 SELECT * FROM Members;
24 CREATE TABLE Loans (
25 loan_id INT PRIMARY KEY,
26 member_id INT,
27 book_id INT,
28 loan_date DATE,
29 FOREIGN KEY (member_id) REFERENCES Members (member_id),
30 FOREIGN KEY (book_id) REFERENCES Books (book_id),
31 );

```

Members 2 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	17:18:39	USE bank_db	0 row(s) affected	0.000 sec
3	17:19:04	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100), author VARCHAR(100), year INT, price DECIMAL(10,2));	0 row(s) affected	0.157 sec
4	17:21:48	INSERT INTO Books (book_id, title, author, year, price) VALUES (1, 'The Great Gatsby', 'F. Scott Fitzgerald', 1925, 10.99), (2, '1984', 'George Orwell', 1949, 12.99), (3, 'The Catcher in the Rye', 'J.D. Salinger', 1951, 10.99);	3 row(s) affected Records: 3 Duplicates: 0 / 0.031 sec	
5	17:24:21	SELECT * FROM Books LIMIT 0, 3;	3 row(s) returned	0.000 sec / 0.000 sec
6	17:29:40	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	0 row(s) affected	0.094 sec
7	17:31:07	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	Error Code: 1050. Table 'members' already exists	0.016 sec
8	17:31:47	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Bindhu Madhavi', 'pennetibindhumadha@gmail.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'Michael Brown', 'michael.brown@email.com');	3 row(s) affected Records: 3 Duplicates: 0 / 0.016 sec	
9	17:33:11	SELECT * FROM Members LIMIT 0, 3;	3 row(s) returned	0.000 sec / 0.000 sec
10	17:34:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_id) REFERENCES Members (member_id), FOREIGN KEY (book_id) REFERENCES Books (book_id));	0 row(s) affected	0.094 sec

Automatic context help disabled. Use the toolbar manually get help for current caret position or toggle automatic help.

Step 9 - Insert Loans

Inserts sample loan transactions that link members to the books they have borrowed.

The screenshot shows the MySQL Workbench interface with the 'Loans' table being populated. The SQL editor contains the following code:

```
27      book_id INT,  
28      loan_date DATE,  
29      FOREIGN KEY (member_id) REFERENCES Members (member_id),  
30      FOREIGN KEY (book_id) REFERENCES Books (book_id)  
31  );  
32  • INSERT INTO Loans (loan_id, member_id, book_id, loan_date)  
33    VALUES  
34    (1, 101, 1, '2025-01-05'),  
35    (2, 102, 2, '2025-01-08'),  
36    (3, 103, 3, '2025-01-10'),  
37    (4, 101, 2, '2025-02-01'),  
38    (5, 102, 1, '2025-02-05'),  
39    (6, 103, 2, '2025-02-12'),  
40    (7, 101, 3, '2025-03-01'),  
41    (8, 102, 3, '2025-03-07'),  
42    (9, 103, 1, '2025-03-15'),  
43    (10, 101, 1, '2025-04-01');
```

The Output tab shows the execution results:

#	Time	Action	Message	Duration / Fec
3	17:19:04	CREATE TABLE Books (boo...	0 row(s) affected	0.157 sec
4	17:21:48	INSERT INTO Books (book_id, ...	3 row(s) affected Records: 3 D...	0.031 sec
5	17:24:21	SELECT * FROM Books LIMIT ...	3 row(s) returned	0.000 sec / 0
6	17:29:40	CREATE TABLE Members (...	0 row(s) affected	0.094 sec
7	17:31:07	CREATE TABLE Members (...	Error Code: 1050. Table 'membe...	0.016 sec
8	17:31:47	INSERT INTO Members (memb...	3 row(s) affected Records: 3 D...	0.016 sec
9	17:33:11	SELECT * FROM Members LIMI...	3 row(s) returned	0.000 sec / 0
10	17:34:58	CREATE TABLE Loans (loan...	0 row(s) affected	0.094 sec
11	17:36:43	INSERT INTO Loans (loan_id, ...	10 row(s) affected Records: 10 ...	0.016 sec

Step 10 - Display Loans

Displays all loan records to confirm the borrowing transactions were recorded.

The screenshot shows the MySQL Workbench interface with the query `SELECT * FROM Loans;` executed. The Result Grid displays the following data:

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07

The Output tab shows the execution results:

#	Time	Action	Message	Duration / Fec
4	17:21:48	INSERT INTO Books (book_id, ...	3 row(s) affected Records: 3 D...	0.031 sec
5	17:24:21	SELECT * FROM Books LIMIT ...	3 row(s) returned	0.000 sec / 0
6	17:29:40	CREATE TABLE Members (...	0 row(s) affected	0.094 sec
7	17:31:07	CREATE TABLE Members (...	Error Code: 1050. Table 'membe...	0.016 sec
8	17:31:47	INSERT INTO Members (memb...	3 row(s) affected Records: 3 D...	0.016 sec
9	17:33:11	SELECT * FROM Members LIMI...	3 row(s) returned	0.000 sec / 0
10	17:34:58	CREATE TABLE Loans (loan...	0 row(s) affected	0.094 sec
11	17:36:43	INSERT INTO Loans (loan_id, ...	10 row(s) affected Records: 10 ...	0.016 sec
12	17:38:21	SELECT * FROM Loans LIMIT ...	10 row(s) returned	0.000 sec / 0

Step 11 - JOIN Query

Joins the Books, Members, and Loans tables to show which member borrowed which book.

Step 12 – Group By Query

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'Navigator' pane with sections for 'MANAGEMENT' (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), 'INSTANCE' (Startup / Shutdown, Server Logs, Options File), 'PERFORMANCE' (Dashboard, Performance Reports), and 'Administration' (Schemas). The main editor displays a SQL query in 'Query 1' (SQL File 3*). The query is a JOIN query that selects member names and book titles from the Loans, Members, and Books tables, joined on member_id and book_id. The 'Result Grid' shows the output of the query, displaying 10 rows of data. The 'Output' pane at the bottom shows the execution log, including the time taken for each query and the number of rows returned.

```
44 SELECT m.full_name AS Member_Name,  
45        b.title AS Book_Title  
46 FROM Loans l  
47 INNER JOIN Members m  
48 ON l.member_id = m.member_id  
49 INNER JOIN Books b  
50 ON l.book_id = b.book_id;
```

Member_Name	Book_Title
Bindhu Madhavi	The Great Gatsby
Bindhu Madhavi	To Kill a Mockingbird
Bindhu Madhavi	1984
Bindhu Madhavi	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984

#	Time	Action	Message	Duration / Fetch
5	17:24:21	SELECT * FROM Books LIMIT ...	3 row(s) returned	0.000 sec / 0.000 se
6	17:29:40	CREATE TABLE Members (...	0 row(s) affected	0.094 sec
7	17:31:07	CREATE TABLE Members (...	Error Code: 1050. Table 'memb...	0.016 sec
8	17:31:47	INSERT INTO Members (memb...	3 row(s) affected Records: 3 D...	0.016 sec
9	17:33:11	SELECT * FROM Members LIM...	3 row(s) returned	0.000 sec / 0.000 se
10	17:34:58	CREATE TABLE Loans (loan...	0 row(s) affected	0.094 sec
11	17:36:43	INSERT INTO Loans (loan_id, ...	10 row(s) affected Records: 10 ...	0.016 sec
12	17:38:21	SELECT * FROM Loans LIMIT ...	10 row(s) returned	0.000 sec / 0.000 se
13	17:39:48	SELECT m.full_name AS Memb...	10 row(s) returned	0.015 sec / 0.000 se

Groups loan records by member to count how many books each member has borrowed.

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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INSTANCE

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 SQL File 3*

```

49 INNER JOIN Books b
50 ON l.book_id = b.book_id;
51 SELECT published_year,
52 COUNT(book_id) AS Total_Books
53 FROM Books
54 GROUP BY published_year
55 ORDER BY published_year;

```

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Result 7 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	17:33:11	SELECT * FROM Members LIM...	3 row(s) returned	0.000 sec / 0.000 sec
10	17:34:58	CREATE TABLE Loans (loan...	0 row(s) affected	0.094 sec
11	17:36:43	INSERT INTO Loans (loan_id, ...	10 row(s) affected Records: 10 ...	0.016 sec
12	17:38:21	SELECT * FROM Loans LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec
13	17:39:48	SELECT m.full_name AS Memb...	10 row(s) returned	0.015 sec / 0.000 sec
14	17:41:13	CREATE INDEX idx_account_t...	Error Code: 1061. Duplicate key...	0.016 sec
15	17:41:13	SELECT * FROM Accounts WH...	2 row(s) returned	0.000 sec / 0.000 sec
16	17:42:26	SELECT m.full_name AS Memb...	10 row(s) returned	0.000 sec / 0.000 sec
17	17:49:53	SELECT published_year, C...	3 row(s) returned	0.000 sec / 0.000 sec

Step 13 - Create Donation History Table

Creates a table to track book donations made by library members.

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 SQL File 3*

```

47 INNER JOIN Members m
48 ON l.member_id = m.member_id
49 INNER JOIN Books b
50 ON l.book_id = b.book_id;
51 SELECT published_year,
52 COUNT(book_id) AS Total_Books
53 FROM Books
54 GROUP BY published_year
55 ORDER BY published_year;
56 CREATE TABLE Donation_History (
57 donation_id INT PRIMARY KEY,
58 book_id INT,
59 donor_name VARCHAR(100),
60 donation_date DATE,
61 FOREIGN KEY (book_id) REFERENCES Books (book_id)
62 );

```

Result Grid

Result 7 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	17:34:58	CREATE TABLE Loans (loan...	0 row(s) affected	0.094 sec
11	17:36:43	INSERT INTO Loans (loan_id, ...	10 row(s) affected Records: 10 ...	0.016 sec
12	17:38:21	SELECT * FROM Loans LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec
13	17:39:48	SELECT m.full_name AS Memb...	10 row(s) returned	0.015 sec / 0.000 sec
14	17:41:13	CREATE INDEX idx_account_t...	Error Code: 1061. Duplicate key...	0.016 sec
15	17:41:13	SELECT * FROM Accounts WH...	2 row(s) returned	0.000 sec / 0.000 sec
16	17:42:26	SELECT m.full_name AS Memb...	10 row(s) returned	0.000 sec / 0.000 sec
17	17:49:53	SELECT published_year, C...	3 row(s) returned	0.000 sec / 0.000 sec
18	17:52:32	CREATE TABLE Donation_Hist...	0 row(s) affected	0.109 sec

Step 14 - Start Transaction

Begin a transaction so that multiple related operations can be executed and committed together.

MySQL Workbench interface showing a SQL query window with the following code:

```
48 ON l.member_id = m.member_id
49 INNER JOIN Books b
50 ON l.book_id = b.book_id;
51 • SELECT published_year,
52     COUNT(book_id) AS Total_Books
53 FROM Books
54 GROUP BY published_year
55 ORDER BY published_year;
56 • CREATE TABLE Donation_History (
57     donation_id INT PRIMARY KEY,
58     book_id INT,
59     donor_name VARCHAR(100),
60     donation_date DATE,
61     FOREIGN KEY (book_id) REFERENCES Books (book_id);
62 );
63 • START TRANSACTION;
```

The Output window shows the following results:

#	Time	Action	Message	Duration / Fetch
11	17:36:43	INSERT INTO Loans (loan_id, ...	10 row(s) affected Records: 10 ...	0.016 sec
12	17:38:21	SELECT * FROM Loans LIMIT ...	10 row(s) returned	0.000 sec / 0.000
13	17:39:48	SELECT m.full_name AS Memb...	10 row(s) returned	0.015 sec / 0.000
14	17:41:13	CREATE INDEX idx_account_t...	Error Code: 1061. Duplicate key...	0.016 sec
15	17:41:13	SELECT * FROM Accounts WH...	2 row(s) returned	0.000 sec / 0.000
16	17:42:26	SELECT m.full_name AS Memb...	10 row(s) returned	0.000 sec / 0.000
17	17:49:53	SELECT published_year, C...	3 row(s) returned	0.000 sec / 0.000
18	17:52:32	CREATE TABLE Donation_Hist...	0 row(s) affected	0.109 sec
19	17:54:23	START TRANSACTION	0 row(s) affected	0.000 sec

Step 15 - Insert New Book

Inserts a new book record as part of the ongoing transaction.

MySQL Workbench interface showing the same SQL query window with the following code:

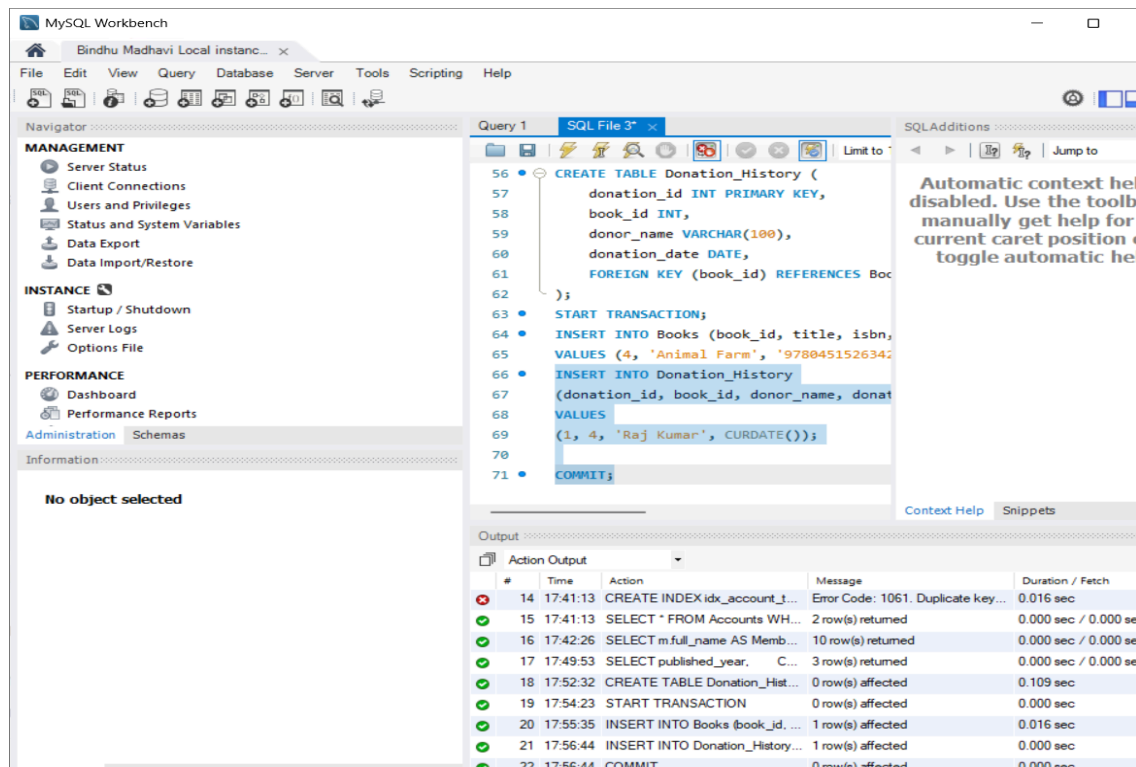
```
50 ON l.book_id = b.book_id;
51 • SELECT published_year,
52     COUNT(book_id) AS Total_Books
53 FROM Books
54 GROUP BY published_year
55 ORDER BY published_year;
56 • CREATE TABLE Donation_History (
57     donation_id INT PRIMARY KEY,
58     book_id INT,
59     donor_name VARCHAR(100),
60     donation_date DATE,
61     FOREIGN KEY (book_id) REFERENCES Books (book_id);
62 );
63 • START TRANSACTION;
64 • INSERT INTO Books (book_id, title, isbn,
65     VALUES (4, 'Animal Farm', '9780451526342
```

The Output window shows the following results:

#	Time	Action	Message	Duration / Fetch
12	17:38:21	SELECT * FROM Loans LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec
13	17:39:48	SELECT m.full_name AS Memb...	10 row(s) returned	0.015 sec / 0.000 sec
14	17:41:13	CREATE INDEX idx_account_t...	Error Code: 1061. Duplicate key...	0.016 sec
15	17:41:13	SELECT * FROM Accounts WH...	2 row(s) returned	0.000 sec / 0.000 sec
16	17:42:26	SELECT m.full_name AS Memb...	10 row(s) returned	0.000 sec / 0.000 sec
17	17:49:53	SELECT published_year, C...	3 row(s) returned	0.000 sec / 0.000 sec
18	17:52:32	CREATE TABLE Donation_Hist...	0 row(s) affected	0.109 sec
19	17:54:23	START TRANSACTION	0 row(s) affected	0.000 sec
20	17:55:35	INSERT INTO Books (book_id, ...	1 row(s) affected	0.016 sec

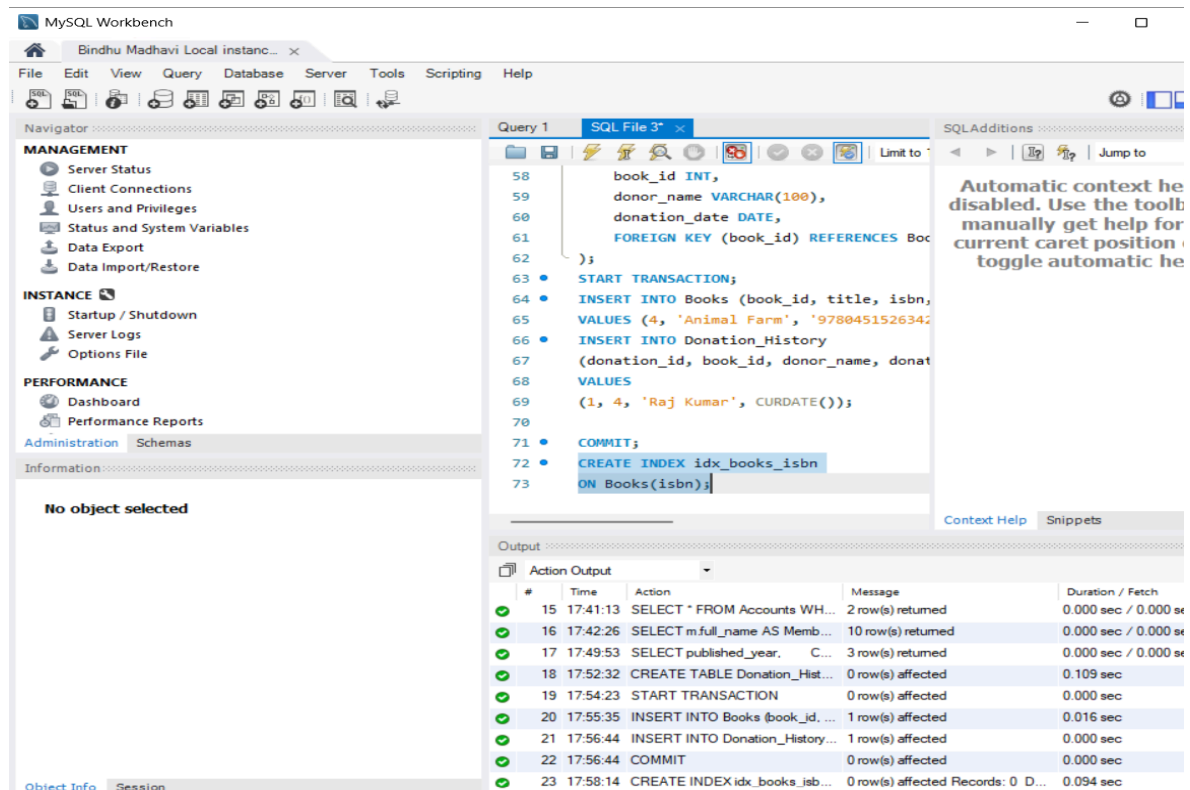
Step 16 - Insert Donation History and Commit

Records a donation entry and commits the transaction to permanently save the changes.



Step 17 - Create Index

Creates an index on a column to speed up future search queries.



Step 18 - Search Using Index

Executes a search query that uses the newly created index for faster retrieval.

MySQL Workbench

Bindhu Madhavi Local instanc...

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Information

No object selected

Query 1 SQL File 3*

```
69 (1, 4, 'Raj Kumar', CURDATE());
70
71 COMMIT;
72 CREATE INDEX idx_books_isbn
73 ON Books(isbn);
74 SELECT * FROM Books
75 WHERE isbn = '9780451524935';
```

SQLAdditions

Automatic context help is disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.

Result Grid

	book_id	title	isbn	published_year
3	1984	9780451524935	1949	
*	NULL	NULL	NULL	NULL

Books 8

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 16	17:42:26	SELECT m.full_name AS Memb...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 17	17:49:53	SELECT published_year, C...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 18	17:52:32	CREATE TABLE Donation_Hist...	0 row(s) affected	0.109 sec
✓ 19	17:54:23	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 20	17:55:35	INSERT INTO Books (book_id, ...	1 row(s) affected	0.016 sec
✓ 21	17:56:44	INSERT INTO Donation_History...	1 row(s) affected	0.000 sec
✓ 22	17:56:44	COMMIT	0 row(s) affected	0.000 sec
✓ 23	17:58:14	CREATE INDEX idx_books_isb...	0 row(s) affected Records: 0 D...	0.094 sec
✓ 24	17:59:33	SELECT * FROM Books WHERE...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session